1 (a) time-dependent

(b) (i) Normalizable (ii) Not normalizable

(c) $e^{ikx}$ is an eigenfunction of $\hat{p}_x^2$ with $\hbar^2 k^2$ as eigenvalue.

2 (a)

$$\hat{A}^2 = \hat{A}$$

$$\hat{A}\psi = \lambda\psi \quad \text{--- (1)}$$

$$\hat{A}(\hat{A}\psi) = \lambda^2\psi \quad \text{--- (2)}$$

$$\hat{A}^2\psi = \lambda^2\psi \quad \text{---}$$

$$\hat{A}\psi = \lambda^2\psi \quad \text{--- (3)}$$

Comparing (1) & (3)

$$(\lambda^2 - \lambda)\psi = 0$$

$\lambda = 0$ or $1$ (proved)

(b)

(i) $\int \psi_a^* \psi_b \, dx = 0$ ; $\int \psi_a^* \psi_a \, dx = \int \psi_b^* \psi_b \, dx = 1$

orthonormal

(ii) $$\langle \hat{A} \rangle = \frac{\int \psi_a \hat{A} \psi_a \, dx}{\int \psi_a^* \psi_a \, dx} = \frac{33}{5}$$

3 (a) $$\boxed{\nabla^2\psi - \frac{e^2}{4\pi\epsilon_0}\psi = E\psi}$$

(b) $[\hat{H}, \hat{L}^2] = 0$ can be shown by ~~until taking the~~ expressing $\hat{H}$ in terms of $\hat{L}^2$.

(c) $\hat{L}^2 Y_l^m(\theta, \phi) = l(l+1)\hbar^2 Y_l^m(\theta, \phi)$

(d) $[\hat{L}_y, \hat{L}_x] = -i\hbar\hat{L}_z$ (Can be proved)

(4) (a) $$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m_p eV}} \simeq 0.082 \text{ volt}$$

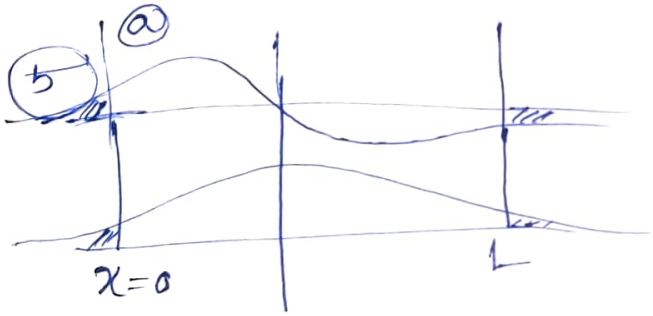
(b) $$\Delta E \, \Delta t \geq \hbar/2$$

$$\Delta E \approx \frac{\hbar}{2 \cdot \Delta t}$$

(c) $$\frac{d\langle \hat{p}_x \rangle}{dt} = \left\langle \frac{\partial \hat{p}_x}{\partial t} \right\rangle + \frac{1}{i\hbar} \left\langle \left[ \hat{p}_x, \frac{\hat{p}_x^2}{2m} + \frac{1}{2} K x^2 \right] \right\rangle$$

$$\frac{d\langle \hat{p}_x \rangle}{dt} = \left\langle \frac{\partial \hat{p}_x}{\partial t} \right\rangle - \langle K \hat{x} \rangle$$

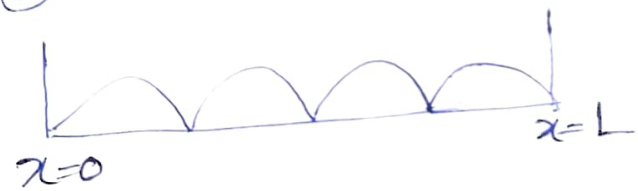
(5) (a)



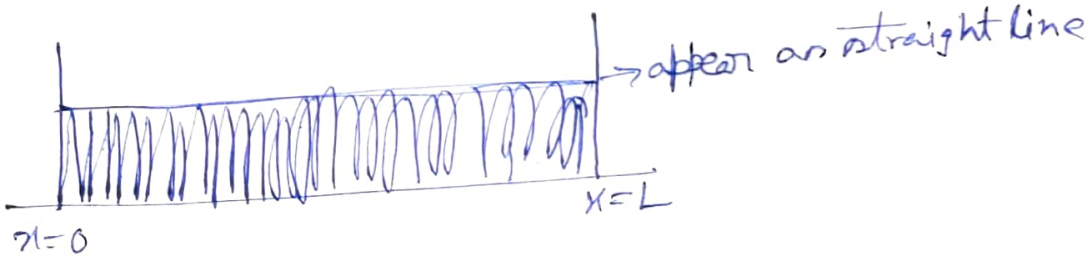

(b) $$\left[ (\hat{p}_x^2 + \hat{p}_y^2), \hat{x} \right] f(x)$$

$$= -2i\hbar \hat{p}_x f(x)$$

(6) (a)




(b)



(c) $\langle x \rangle = \left(\frac{2}{L}\right)\int_0^L x \sin^2 \frac{n\pi x}{L}\, dx = \frac{L}{2}$ for all $n$

7. (a)

$$H = -\frac{\hbar^2}{2m}\left[\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}\right]+\frac{1}{2}m\omega_x^2+\frac{1}{2}m\omega_y^2+\frac{1}{2}m\omega_z^2$$

(b)

$$E_{n_x n_y n_z} = \left(n_x+\tfrac{1}{2}\right)\hbar\omega_x+\left(n_y+\tfrac{1}{2}\right)\hbar\omega_y+\left(n_z+\tfrac{1}{2}\right)\hbar\omega_z$$

(c)

$$E_{0,0,0} = \tfrac{1}{2}\hbar\omega_x+\tfrac{1}{2}\hbar\omega_y+\tfrac{1}{2}\hbar\omega_z$$

(d)

if $\omega_x=\omega_y=\omega_z=\omega$

first excited state $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ triply degenerate

$$E_{1,00} = E_{010} = E_{001} = \left(\tfrac{3}{2}+1\right)\hbar\omega$$
$$= \tfrac{5}{2}\hbar\omega$$

(e)

(i) $K = m\omega^2$

$K = m(2\pi\nu)^2$ force constant

(ii) $(n+\tfrac{1}{2})\hbar\omega = \tfrac{1}{2}m\omega^2 A^2$

$A$ = amplitude = 4 cm

Solving $n$ = Corresponding quantum number
= very large

(8) $n = 2$ $\quad n^2 = 2^2 = 4$ (degenerate)

(a) $l = 0, 1$

for $l = 0$ $m = 0$,

$l = 1$ $m = 0, +1$

$$\left.\begin{array}{l} R_{20}\, Y_0^0(\theta,\phi) \\ R_{21}\, Y_1^0(\theta,\phi) \\ R_{21}\, Y_1^{+1}(\theta,\phi) \end{array}\right\}$$ all degenerate eigenfunctions.

(9) (a)

$$E = \frac{l(l+1)\hbar^2}{2\mu r^2}$$

$E_{l+1}$ to $E_l$ transition

$$\Delta E = \left[\frac{(l+1)(l+2)\hbar^2}{2\mu r^2} - \frac{l(l+1)\hbar^2}{2\mu r^2}\right]$$

$$= \frac{(l+1)\hbar^2}{\cancel{2}\mu r^2}\left[\cancel{2}\right]$$

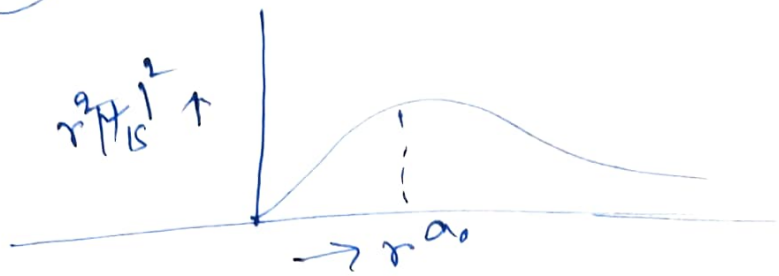
$$= \frac{(l+1)\hbar^2}{\mu r^2}$$

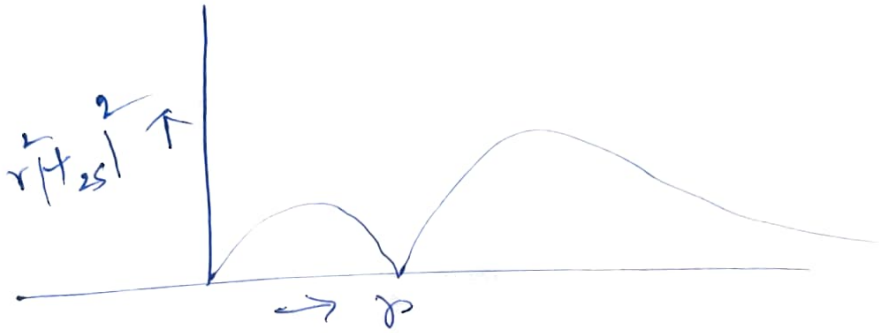
for $l \to 0$ to $1$

$$\Delta E = \frac{\hbar^2}{\mu r^2}$$

$$r = \left(\frac{\hbar^2}{\mu \Delta E}\right)^{1/2}$$

(b)




(c)

by putting $\left.\dfrac{d(r^2|\psi_{1s}|^2)}{dr}\right] = 0$

get $r = a_0$

(d) $E_n = -\dfrac{Z^2\, 13.6\ eV}{n^2}$ ($n=3$ all cases)

for $He^+$ $Z=2$

$Li^{2+}$, $Z=3$

$Be^{3+}$ $Z=4$